STAT 325 Homework 3. 2/5/26.

1. $N = 65 \quad a = 5 \rightarrow N/a = n = 13$

	Control	1	2	3	4
$\bar{y}_{i\cdot}$	11.54	11.00	11.42	11.44	11.28
S_i	0.27	0.47	0.31	0.42	0.31

$$SS_{Treatment} = \left(\frac{1}{n}\sum y_{i\cdot}^2\right) - \frac{y_{\cdot\cdot}^2}{N}$$

$$y_{\cdot\cdot} = 13(11.54 + 11 + 11.42 + 11.44 + 11.28) = 736.84$$

$$y_{\cdot\cdot} = n\left(\sum_{j=1}^{a} \bar{y}_{i\cdot}\right)$$

$$\frac{y_{\cdot\cdot}^2}{N} = \frac{(736.84)^2}{65} = 8352.8182$$

$$\frac{\sum(y_{i\cdot}^2)}{n} = \frac{\sum(\bar{y}_{i\cdot}n)^2}{n} = n\sum_{j=1}^{a}\bar{y}_{i\cdot}^2 = 13(11.54^2 + 11^2 + 11.42^2 + 11.44^2 + 11.28^2)$$

$$= 8355.1$$

$$SS_{Trt} = 8355.1 - 8352.8182 = 2.2817$$

$$SS_E = \sum_{i=1}^{a}\sum_{j=1}^{13} y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N} = \sum_{i=1}^{a}(n-1)S_i^2 = \left\{\begin{array}{l} 12(0.27)^2 = 0.8748 \\ 12(0.47)^2 = 2.6508 \\ 12(0.31)^2 = 1.1532 \\ 12(0.42)^2 = 2.1168 \\ 12(0.31)^2 = 1.1532 \end{array}\right. +$$

$$\approx 7.9488$$

$$SS_T = SS_E + SS_{trt} = 2.2817 + 7.9488$$

$$= 10.2305$$

$$F_0 = \frac{MS_{Betw}}{MS_{within}} = \frac{0.5705}{0.1324}$$

$F_{\alpha=0.01}$ 4.60

Variation	SS	df	MS	F_0	P
Between	2.282	4	$2.282/4 \approx 0.5705$	≈ 4.3063	0.00396
within	7.9488	60	$7.9488/60 \approx 0.13248$		
Total	10.2305	64			

R: $1 - pf(F_0$, $df_1 = 4$, df_2

(2) $E(MS_{trt}) = E\left(\frac{SS_{trt}}{a-1}\right) = E\left(\frac{1}{a-1}\left(\frac{1}{n}\sum y_{i\cdot}^2 - \frac{y_{\cdot\cdot}^2}{N}\right)\right)$

We assume the effect model

$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ so plug in

$$= \frac{1}{a-1} E\left[\frac{1}{n}\sum_{i=1}^{a}\left(\sum_{j=1}^{n}\mu + \tau_i + \varepsilon_{ij}\right)^2 - \frac{1}{N}\left(\sum_i\sum_j \mu + \tau_i + \varepsilon_{ij}\right)^2\right]$$

$\sum_i \tau_i = 0$

~~$\frac{1}{a-1} E\left(\frac{1}{n}\left(an\mu + n\sum_{j=1}^{a}\tau_i\right.\right.$~~

$$= \frac{1}{a-1} E\left(\frac{1}{n}\sum_{i=1}^{a}\left(n\mu + n\tau_i + \sum_{j=1}^{n}\varepsilon_{ij}\right)^2 - \frac{1}{N}\left(na\mu + n\sum_i^a \tau_i + \sum_i\sum_j \varepsilon_{ij}\right)^2\right)$$

$$= \frac{1}{a-1} E\left(\frac{1}{n}\sum_i \left(n^2\mu^2 + n^2\tau_i^2 + \varepsilon_{i\cdot}^2 + 2n^2\mu\tau_i + 2n\mu\varepsilon_{i\cdot} + 2n\tau_i\varepsilon_{i\cdot}\right)\right.$$

$$\left. - \frac{1}{N}\left(N\mu + \varepsilon_{\cdot\cdot}\right)^2\right)$$

$$= \frac{1}{a-1} E\left(\left(n\mu^2 a + n\sum_{i=1}^{a}\tau_i^2 + \frac{1}{n}\sum_{i}^{a}\varepsilon_{i\cdot}^2 + 2n\mu\underbrace{\sum_i^a \tau_i}_{0}\right.\right.$$

$$\left.\left. + 2\mu\sum_{i=1}^{a}\varepsilon_{i\cdot} + 2\underbrace{\sum_i \tau_i\varepsilon_{i\cdot}}_{0}\right) - \frac{1}{N}\left(N^2\mu^2 + \varepsilon_{\cdot\cdot}^2 + 2N\mu\varepsilon_{\cdot\cdot}\right)\right)$$

$$= \frac{1}{a-1} E\left(\left(an\mu^2 + n\sum_i^a \tau_i^2 + \frac{1}{n}\sum_i \varepsilon_{i\cdot}^2 + 2\mu\varepsilon_{\cdot\cdot} + 2\sum_i^a \tau_i\varepsilon_{i\cdot}\right)\right.$$

$$\left. - \left(N\mu^2 + 2\mu\varepsilon_{\cdot\cdot} + \frac{\varepsilon_{\cdot\cdot}^2}{N}\right)\right)$$ (Take expected value)

$$= \frac{\sigma_\tau^2 + a\sigma^2 - \sigma^2}{a-1} = \frac{(a-1)\,\sigma_\tau^2 + (a-1)\,\sigma^2}{a-1} = \sigma^2 + n\sigma_\tau^2$$